

The impact of uncorrected astigmatism on night driving performance

Alex A Black , Joanne M Wood , Luisa H Colorado  and Michael J Collins 

School of Optometry and Vision Science and the Institute of Health and Biomedical Innovation, Queensland University of Technology, Brisbane, Australia

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Correspondence: Alex A Black
E-mail address: aa.black@qut.edu.au

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Abstract

Purpose: To investigate the effect of uncorrected astigmatism on night driving performance on a closed-road circuit.

Methods: Participants included 10 drivers (mean age 24.4 ± 7.0 years), with low to moderate bilateral astigmatism (0.75–1.50 DC), who were regular contact lens (CL) wearers. Vision and night driving performance were assessed in a cross-over design with a toric CL and a best-sphere spherical CL. Binocular visual function measures included photopic and mesopic visual acuity (VA), contrast sensitivity (CS), mesopic motion sensitivity and glare tests (Mesotest[®] II and Halometer). Night-time driving performance was assessed on a closed-road circuit, which included measures of sign recognition, hazard detection and avoidance, pedestrian recognition distances, lane keeping, speed and overall driving score.

Results: Correction of astigmatism with toric CL significantly improved mesopic VA, photopic and mesopic CS, mesopic motion sensitivity, and reduced glare ($p < 0.05$), compared to the spherical CL; there were no significant effects of visual correction type on photopic VA. Correction of astigmatism using toric CL resulted in significant improvements in night driving performance, compared to driving with spherical CL, particularly for sign recognition, avoidance of low contrast hazards, increased pedestrian recognition distances and overall driving score ($p < 0.05$).

Conclusions: Correction of low to moderate levels of astigmatism had significant positive effects on night-time driving performance and related tests of visual performance. This has important implications for optical corrections to improve night road safety of drivers with astigmatism.

Background

Low levels of astigmatism are common, with between 47% and 51% of patients presenting for clinical care having at least 0.75 D astigmatism in one eye, while around 35% have at least 1.00 D of astigmatism in one eye.^{1,2} However the rate of correction of astigmatism with soft toric contact lenses (CL) is comparatively low, ranging from about 10% to 40% of soft CL wearers across various countries.^{3,4} A variety of factors may account for the low prescribing rate of soft toric CLs, including the relative complexity of toric CL fitting, comfort differences between spherical and toric CL, varying patient tolerance of visual compromise, and

cost.⁵ A considerable proportion of low-to-moderate astigmatic patients are thus likely to have their astigmatism intentionally uncorrected when wearing CL, which may impact on their ability to perform everyday tasks such as reading, computer work and driving.

The detrimental effect of uncorrected astigmatism on standard vision measures such as visual acuity and contrast sensitivity are well known.^{6,7} However, little is known about its impact on other visual function measures, such as glare sensitivity and visual function under mesopic conditions. One study demonstrated that toric CL correction in individuals with ≤ 1.00 D of astigmatism, improved high and low contrast visual acuity by

around three letters under photopic conditions, and four to five letters under mesopic conditions ($\leq 3 \text{ cd m}^{-2}$) compared to spherical correction.⁸

Importantly, little is known about the impact of astigmatism on functional measures such as driving. Only two studies have explored the effects of astigmatic blur on driving performance in simulators and reported no significant impact of either induced or uncorrected astigmatism on driving performance or reaction times, when compared to best-corrected vision.^{9,10} However, both studies were undertaken in driving simulators under daytime conditions, which have been shown to be less affected by blur than night-time driving.¹¹ In addition, driving simulators do not capture the lighting conditions, complexity and challenges of real-world driving.¹²

Indeed, driving under night-time conditions is more challenging than in the daytime. Fatality rates from road crashes are up to three times higher than for daytime driving when adjusted for distances driven,¹³ particularly for crashes involving pedestrians,¹⁴ with crashes often attributed to poor visibility at night.^{15,16} Closed-road studies have demonstrated that driving at night with spherical refractive blur, with levels as low as +0.50 DS, result in a significant reduction in overall driving performance, as well as pedestrian recognition distances compared to best-corrected vision.^{11,17} However, there has been limited research into the effect of uncorrected astigmatism on driving performance at night.

The purpose of this study was to test the hypothesis that correction of low-to-moderate astigmatism using toric CL would improve night driving performance, as measured under standardised conditions on a closed-road circuit, compared to correction with spherical CL.

Methods

Participants

Ten visually normal adults (mean age 24.4 ± 7.0 years; range 19–39 years; nine females, one male) who were existing spherical soft contact lens wearers (either occasional or regular wearers) were recruited from the Queensland University of Technology Optometry Clinic and a database of research participants. All participants were regular drivers and currently licensed in Australia, and had not participated in any previous driving research on the closed-road circuit.

Participants were required to have monocular best spectacle-corrected visual acuity of 0.00 logMAR (Snellen 6/6) or better, spherical refraction between 0.00 and -6.00 dioptres (D) and cylinder refraction between -0.75 to -1.50 D in both eyes (any axis). Participants were excluded if they had a history of ocular surgery/trauma, presence of ocular diseases known to affect contact lens

(CL) wear, use of any medications known to interfere with CL wear and pregnant women.

The study followed the tenets of the Declaration of Helsinki and was approved by the Queensland University of Technology Human Research Ethics Committee. All participants were given a full explanation of the nature of the study and experimental procedures and possible consequences of the study, and written informed consent was obtained. The study was a double-masked, randomised, cross-over study, which was conducted over two sessions: one session for contact lens fitting and visual function testing, followed by a night-time driving assessment.

Driving characteristics

A custom-designed questionnaire was self-administered to characterise the driving habits of participants and ratings of vision with habitual optical correction while driving during the day and at night. The Vision and Night Driving Questionnaire (VND-Q) was used to assess participants' vision-related night driving difficulties with their habitual optical correction.¹⁸ The VND-Q contains nine items scored on a 5-point scale (1 = 'No difficulty' to 5 = 'Extreme difficulty') across a range of common night driving situations, such as seeing the road in the rain or poor weather, or seeing pedestrians or animals on the roadside. Scores were converted to Rasch-scaled, interval-level scores (from 0 to 10), where higher scores represent more difficulties.

Contact lens characteristics and fitting

Before CL fitting, participants underwent a slit-lamp biomicroscopic examination and corneal topography using the Medmont E300 videokeratoscope (www.medmont.com). Monocular spherocylindrical and best sphere refractions were undertaken under dim illumination (15 ± 5 lux) to determine the CL prescriptions, using a strategy of maximising plus power without any detriment to visual acuity. Participants were fitted with two disposable lens types: a two-weekly replacement toric soft lens (Johnson and Johnson Vision Care, <https://www.jnjvisioncare.se>, Acuvue Oasys[®] Brand Contact Lenses for Astigmatism) and a two-weekly replacement spherical soft lens (Johnson and Johnson Vision Care, Acuvue Oasys[®] Brand Contact Lenses).

For the toric CL prescription, a -0.75 D cylinder was prescribed when the refraction was -0.75 or -1.00 DC, and a -1.25 D cylinder prescribed when the refraction was -1.25 DC or -1.50 DC, with an 8.6 mm base curve (lens diameter 14.5 mm). The nearest available cylinder axis was used (10 degree steps). For the spherical CL prescription, an 8.4 or 8.8 mm base curve was used (lens diameter 14.0 mm for both base curves), based on corneal

topographical assessment and a best-sphere over-refraction was carried out. Lens fits were assessed after a 15-min settling period with assessment of full corneal coverage, optimal push-up, and stable toric lens rotation with the cylinder axis within 10 degrees from the optimal location. Final optimal refraction with stable visual acuity was applied to all fits.

Visual function assessment

Visual function was assessed for participants wearing each of the study lenses, after a 10-min settling period; testing order was counter-balanced between participants.

Visual acuity

Binocular distance visual acuity was tested using Early Treatment Diabetic Retinopathy Study (ETDRS) charts at 4 m under two lighting conditions: photopic (108 cd m^{-2}) under normal room lighting (440 lux; vertical at the eye) and mesopic (0.75 cd m^{-2}) under dim room lighting (5 lux; vertical at the eye). Visual acuity was scored letter by letter, in logMAR, with a termination rule of stopping after three or more mistakes on a line. Four different charts were used for each light level and lens type.

Contrast sensitivity

Binocular contrast sensitivity was measured with the Pelli-Robson chart at 3 m under the normal and mesopic lighting conditions. Contrast sensitivity was scored letter by letter, in logCS. Two different charts were used for each light level.

Mesotest

Binocular mesopic contrast sensitivity (without and with glare) was tested using the OCULUS Mesotest® II (www.oculus.de/en). The test involves detection of the orientation of a 20/200 (6/60) Landolt C (six-alternate forced-choice) at a virtual distance of 5 m, where target contrast varied by 0.1 log units (95%, 80%, 63% and 50% of the contrast threshold). Five presentations were tested at each contrast level, without and with the glare source, comprising a white LED glare source (0.35 lux at the pupil) located 3 degrees to the left of the target. Background luminance was 0.032 cd m^{-2} without the glare source and 0.1 cd m^{-2} with the glare source. The total number of correctly identified targets was calculated (maximum of 40).

Halometry

The halo produced by glare from a bright white LED attached to the centre of an iPad4 screen (www.apple.com) was determined as the position closest to the LED where two out of three presentations of a randomly presented

high contrast 20/50 (6/15) letter (0.4 logMAR) were correctly identified.¹⁹ A seen-to-not-seen approach was used with a 0.1° step size along 8 meridians. The halo area (cm^2) was determined by calculating the area of the halo surrounding the LED glare source.

Mesopic motion sensitivity

Central motion sensitivity was measured using a computer-based random dot kinematogram (dot density 0.43%, screen luminance $0.36 \pm 0.02 \text{ cd m}^{-2}$) at a distance of 3 m.^{20–22} Participants identified the direction of movement of a central panel of dots that moved randomly in one of four directions and the minimum dot displacement threshold detected (Dmin; log deg arc) was determined using the mean of the last six reversals of a two-down one-up staircase algorithm.

Driving assessment

Night-time driving performance was measured on a closed-road circuit as used in previous studies.^{17,23} Testing was conducted after nautical twilight and when road surfaces were dry. The circuit is 6.8 km long, and included hills, curves, intersections, and straight sections, and did not contain any street lighting.

All driving assessments were conducted in an automatic transmission sedan (Toyota Camry 2015) with the tungsten-halogen headlights set to low-beam for testing, to reflect general driving conditions. The vehicle was instrumented with two roof-mounted cameras, (GoPro HERO3, www.gopro.com) which recorded lane position, and a MUVI HD internal recording device (www.veho-muvi.com) recorded drivers' verbal responses. An in-vehicle GPS logging system recorded the location of the car on the circuit when drivers pressed a response pad.

Each participant completed three laps of the test circuit, including one practice lap and two test laps. The practice lap was used to familiarise the driver with the test vehicle, the driving circuit, and also the tasks required of them. The practice lap was performed in the opposite direction to the test laps, to reduce any practice effects.

The two test laps were conducted with each of the study lenses inserted, after 10 min lens settling time (*Figure 1*). An unmasked investigator (who was not involved in the driving assessment) inserted the study lenses to ensure that participants and driving experimenters were masked to lens type. The two test laps assessed the same components of driving performance but varied the order in which pedestrians were present on the roadside and the location of half of the on-road hazards, to minimise any potential learning effects. The testing order for the two CL conditions (sphere vs toric) was randomised and counter-balanced across participants.

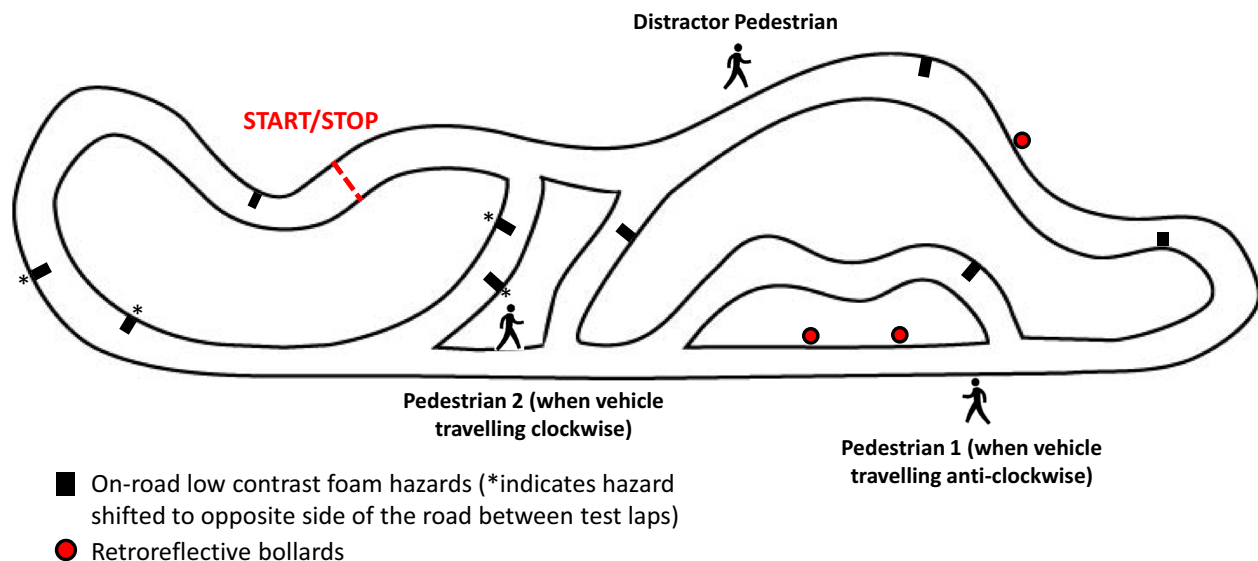


Figure 1. Schematic diagram of the layout of the driving circuit.

Driving performance was assessed using real-world tasks such as reading road signs, recognising and avoiding road hazards, detecting pedestrians, and maintaining lane control. Measures included: (1) sign recognition; (2) hazard avoidance; (3) pedestrian recognition distances; (4) lane keeping; and (5) driving time.

Sign recognition

Participants were instructed to drive at a comfortable speed and verbally report information from road signs that contained 128 items of information. The percentage of sign items correctly reported was calculated.

Hazard avoidance

Participants were required to report and avoid hitting any of 16 large, low-contrast grey foam 'hazards' (220 × 80 × 15 cm) positioned orthogonally in the driving lane along the roadway, the locations of which were altered between laps. The percentage of hazards hit was recorded.

Pedestrian recognition distance

Two experimenters acted as pedestrians and 'walked in-place' at different locations of a 400 m straight section of the circuit. These pedestrians, known as Pedestrian 1 and Pedestrian 2, were not surrounded by any visual clutter or lighting and wore biomotion reflective strips on the moveable joints.^{17,24} On each lap, pedestrians walked in place on the opposite side of the road facing directly towards the oncoming vehicle, which allows for inclusion of naturalistic motion and ensured pedestrian safety. The vehicle travelled this section of the circuit several times in both directions

during each lap, but a pedestrian was present on the road-side only once for each direction (pedestrian 1, vehicle travelling anti-clockwise direction; pedestrian 2, vehicle travelling clockwise direction). Pedestrian recognition distances were calculated from the known GPS coordinates of pedestrians and the GPS coordinates of the car when drivers tapped the touchpad when they recognised pedestrians.

To reduce drivers' expectations that a pedestrian would always be located along the straight stretch of the circuit, an additional distractor pedestrian wearing a retroreflective vest walked in place at a different location of the circuit; data for this pedestrian were not recorded due to the limited sight distances available. In addition, several retroreflective bollards were positioned around the circuit to generate clutter to minimise expectation effects.

Lane keeping

Lane keeping was scored using video playback of the roof-mounted cameras to calculate the percent of total runtime spent driving within the lane markings, excluding lane changes to avoid the road hazards.

Overall driving score

A composite driving score was derived to capture the overall driving performance of each participant compared to the group, as used in previous studies.^{11,23,25,26} The overall score was calculated based on the mean z-scores of the five component driving measures (sign recognition, recognition distances for Pedestrian 1 and Pedestrian 2, hazard avoidance and lane keeping), where a positive z-score represented better performance.

Driving time

The total time taken to complete each run was recorded.

Statistical analysis

Statistical analyses were performed using SPSS statistical software v23.0 (www.ibm.com), and the level of significance was set at $p < 0.05$. Descriptive statistics were used to analyse the demographic, questionnaire and driving characteristics data. The Kolmogorov–Smirnov test was used to check for normality of the distribution of the continuous vision variables. Paired t -tests or Wilcoxon rank-sum, as indicated, were used to compare differences in visual function tests between the two CL conditions (sphere vs toric).

Repeated measures regression analysis using generalised estimating equations (GEE) with a normal link function, with an exchangeable correlation matrix, were performed to assess differences in driving outcome measures between the two lens conditions (sphere vs toric). The driving outcome measures analysed included sign recognition, road hazard avoidance, lane-keeping, pedestrian recognition distance, and the overall driving score. All models included run order and lap time as covariates, to adjust for any potential learning effects and differences due to driving speed. To explore for potential visual predictors of overall night driving performance, each vision measure was entered as a predictor for overall driving score in separate GEE models combined across the two conditions, adjusting for run order and lap time. For all GEE models, residuals were checked for normality and outliers.

Results

Participants mean manifest refraction was -3.14 ± 1.58 D (range -0.75 to -5.50 D) of sphere and -1.04 ± 0.30 D (range -0.75 to -1.50 D) of cylinder. Participants had

5.7 ± 6.7 years of driving experience, and drove on average 3.6 ± 2.0 days per week and 3.3 ± 1.9 nights per week. During the day, all participants reported excellent to good vision when driving. At night, 60% ($n = 6$) reported excellent to good vision, and 40% ($n = 4$) reported fair vision when driving. The mean Rasch-scaled VND-Q score was 3.1 ± 0.7 out of possible 10, indicating relatively low levels of vision-related night driving difficulties.¹⁸

Visual function

The mean visual function for the contact lens conditions are summarised in Table 1. There was no significant difference in photopic visual acuity between the spherical and toric lens conditions ($p = 0.34$). However, mesopic visual acuity was significantly better for the toric compared to spherical correction ($p < 0.001$), with a difference of almost one line of letters. Contrast sensitivity, under both photopic and mesopic light levels, was significantly better for the toric compared to spherical correction ($p < 0.008$), by around four letters.

Participants performed significantly better on the other mesopic and glare visual function measures with the toric compared to spherical correction. This included more letters seen on the Mesotest[®] II, a smaller halometer glare area and higher levels of motion sensitivity ($p < 0.04$).

Night driving performance

There was a significant main effect of correction on the overall driving score, with scores significantly higher (reflecting better performance) for the toric compared to the spherical condition ($p = 0.003$; Figure 2). There were also significant differences for sign recognition ($p = 0.02$), low contrast hazards hit ($p = 0.02$) and the distance at which one pedestrian (Pedestrian 1) was first recognised

Table 1. Visual function performance for the spherical and toric contact lens (CL) corrections

Vision function tests	Spherical CL Mean (S.D.)	Toric CL Mean (S.D.)	Statistic; p value [†]
Spherical prescription, D, mean (S.D.)	RE -3.68 (1.66)LE -3.55 (1.60)	RE -3.13 (1.60)LE -3.03 (1.58)	
Cylinder prescription, D, mean (S.D.)		RE -0.90 (0.24)LE -0.90 (0.24)	
Photopic visual acuity (logMAR), mean (S.D.)	-0.09 (0.08)	-0.11 (0.09)	$t_9 = 1.04$; $p = 0.34$
Mesopic visual acuity (logMAR), mean (S.D.)	0.14 (0.08)	0.05 (0.07)	$t_9 = 5.60$; $p < 0.001^*$
Photopic contrast sensitivity (log units), mean (S.D.)	1.76 (0.03)	1.94 (0.04)	$t_9 = -6.41$; $p < 0.001^*$
Mesopic contrast sensitivity (log units), mean (S.D.)	1.49 (0.21)	1.71 (0.19)	$t_9 = -3.35$; $p = 0.008^*$
Mesotest II Score (# correct out of 40), median (IQR)	38 (37–39)	40 (40–40)	$Z = 26.5$; $p = 0.034^*$
Halometer glare area (cm ²), mean (S.D.)	1.87 (1.13)	1.07 (0.74)	$t_9 = 3.43$; $p = 0.008^*$
Mesopic motion sensitivity (log deg arc), median (IQR) [‡]	-1.82 (-1.82 to -1.74)	-1.93 (-2.00 to -1.82)	$Z = 2.0$; $p = 0.024^*$

IQR, Interquartile range; S.D., Standard Deviation.

* $p < 0.05$.

[†]Paired t -Tests or Wilcoxon Signed Rank tests.

[‡]Lower value reflects better motion sensitivity.

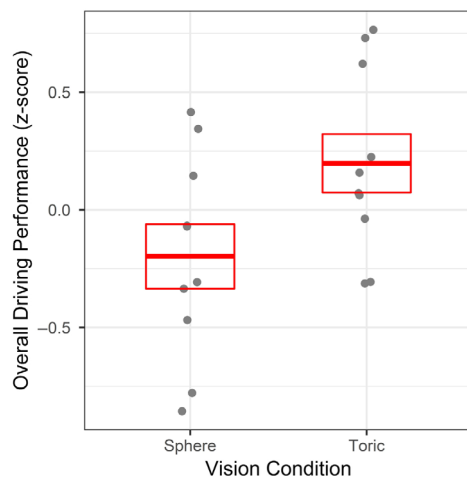


Figure 2. Overall night driving performance with spherical and toric contact lens (CL) correction. The red boxes indicate the mean and ± 1 standard error of mean (S.E.M.). More positive z-score reflects better driving performance.

($p = 0.013$); in all cases, performance was better when participants were wearing toric correction compared to the spherical correction (Table 2). There were no significant differences for the distance at which Pedestrian 2 was recognised ($p = 0.49$), lane-keeping ability ($p = 0.67$) or lap time ($p = 0.72$).

In the models examining the visual predictors of overall driving score for the two vision conditions, better night driving performance was most strongly associated with more targets correctly identified on the Mesotest[®] II ($\chi^2_1 = 14.5$, $p < 0.001$), better mesopic CS ($\chi^2_1 = 11.9$, $p = 0.001$) and smaller halo area ($\chi^2_1 = 11.3$, $p = 0.001$).

Other significant visual predictors were identified, with better night driving performance associated with better visual performance for motion sensitivity ($\chi^2_1 = 9.5$, $p = 0.002$), mesopic VA ($\chi^2_1 = 9.4$, $p = 0.002$), photopic CS ($\chi^2_1 = 9.7$, $p = 0.002$) and photopic VA ($\chi^2_1 = 6.7$, $p = 0.01$).

Discussion

In this study, drivers with low to moderate levels of bilateral astigmatism demonstrated significantly better night-time driving performance when driving with astigmatism corrected using a toric CL correction, compared to a spherical CL correction. This is the first study to demonstrate this relationship using participants with true astigmatic corrections in a real-world night driving environment. Importantly, the findings contribute to understanding the functional effects of uncorrected astigmatism on driving in the visually-challenging night-time road environment, and highlights the benefits of correcting low to moderate astigmatism with toric lens designs for CL wearers who drive at night.

The correction of astigmatism significantly improved overall night driving performance, particularly sign recognition, detection and avoidance of low contrast hazards and recognition distance for one of the two pedestrians. Conversely, two previous studies failed to show any effect of uncorrected astigmatism on driving performance as measured in a driving simulator.^{9,10} However, these studies only assessed driving metrics related to vehicle control (speed, steering, braking, lane keeping), and our study similarly found no differences for lap time (speed) or lane-keeping. Importantly, the driving simulator studies were undertaken under daytime conditions. In a real-world

Table 2. Night driving performance for the spherical and toric contact lens (CL) corrections

Driving variable	Spherical CL		Toric CL		Wald Chi-Square Values (df = 1)	p Value*
	Mean $\pm$ S.D.	Range	Mean (S.D.)	Range		
Overall driving score (z-score), mean (S.D.) [†]	-0.20 ± 0.43	-0.86 to 0.42	0.20 ± 0.39	-0.31 to 0.77	9.11	0.003*
Component driving tasks						
Sign recognition, % seen, mean (S.D.)	61.0 ± 8.1	45.3 to 71.9	65.0 ± 7.7	56.6 to 76.1	5.34	0.021*
Low contrast hazard detection, % hit, median (IQR)	12.9 (6.3 to 38.4)	0 to 84.6	7.6 (0.0 to 20.0)	0 to 23.1	5.25	0.022*
Pedestrian 1 recognition distance, m, mean (S.D.)	96.8 ± 45.2	30.0 to 188.6	116.6 ± 41.2	58.2 to 186.1	7.00	0.008*
Pedestrian 2 recognition distance, m, mean (S.D.)	131.9 ± 56.3	58.5 to 216.8	147.4 ± 41.9	102.9 to 220.7	0.48	0.49
Lane keeping, % of time outside lane, median (IQR)	5.7 (5.2 to 14.3)	2.5 to 21.9	5.2 (4.4 to 16.9)	2.3 to 19.2	0.19	0.67
Lap time, min, mean (S.D.) [‡]	12.1 ± 2.3	9.5 to 16.4	11.6 ± 1.6	9.0 to 14.1	0.13	0.72

* $p < 0.05$.

[†]Adjusted for lap time, and run order.

[‡]Adjusted for run order.

night driving environment, driving performance remained unaffected for up to +1.00 D of spherical blur under day-time conditions, whereas +0.50 DS of spherical blur caused significant decrements in aspects of night-time driving performance.¹¹

Detection and avoidance of low contrast hazards was particularly affected in the presence of uncorrected astigmatism (sphere: 13% vs toric: 8% hazards hit), as was proportion of signs recognised (sphere: 61% vs toric: 65%). The findings of the current study support previous studies showing that low levels of positive spherical lens blur increased the number of hazards hit and reduced the number of signs recognised, as well as overall driving score.¹¹

Pedestrian recognition distances were significantly reduced in the presence of uncorrected astigmatism (around 20 m difference for Pedestrian 1), which again is in accord with our previous studies of night-time pedestrian recognition and refractive blur.¹⁷ However, no significant differences were found for the recognition distances for Pedestrian 2, which may be due the pedestrian at this location being directly within the car headlight beam and more visible, thus affording much longer sighting distances (up to 220 m) which were less affected by uncorrected astigmatism. Overall, these findings support previous studies that showed that low levels of simulated spherical blur reduced pedestrian recognition distances at night, even when pedestrians were wearing clothing configurations incorporating retroreflective strips on the moveable joints.¹⁷ It is possible that these differences may be even greater for pedestrians without retro-reflective material. These findings have important safety implications given the negative impact of uncorrected astigmatism on recognition distances, and the road safety challenges for vulnerable road users such as pedestrians and cyclists at night-time.²⁷

This study demonstrated that visual function measures conducted under mesopic light levels, and photopic contrast sensitivity, were impaired to a greater extent by uncorrected astigmatism than standard photopic measures of visual acuity. This finding may be due to the increase in pupil size under mesopic conditions which increases the retinal blur circle.^{28,29} In addition, performance on the glare tests improved when astigmatism was corrected. These findings have important implications for the assessment of visual functional abilities on night-time driving which should be undertaken under conditions that reflect the night-driving environment.

The study findings need to be considered in light of its strengths and limitations. Strengths include the assessment of participants with true, rather than simulated, astigmatism and the use of a real night-time driving environment. Variations in individual driving performance were controlled through the use of the repeated measures cross-over study design, which included participant-masking and

randomisation to minimise learning effects. Limitations of the study include a relatively small sample size; nevertheless, significant effects were found for a number of driving outcome measures, similar to the effects of simulated spherical blur under night-time conditions in previous studies. In addition, the closed-road circuit was free from other road users; therefore lacked the complexity of real-world in-traffic conditions, and testing was conducted in dry weather conditions. It is likely that the effect of uncorrected astigmatism would be even greater in more complex driving situations, particularly in poor weather conditions at night. Future studies need to explore these effects for a larger sample size, with a wider range of cylindrical powers, and to explore whether differences in driving performance exist dependent on the direction of the astigmatic axis.

In summary, this study demonstrated that the correction of low to moderate levels of astigmatism with toric CL, compared to spherical CL, provides significant improvements for visual function measures relevant to night driving, and most importantly, night-time driving performance, including hazard detection and recognition of signs and pedestrians. The results highlight the benefits for road safety of optimally correcting patients with low to moderate astigmatism when driving at night and the utility of visual function tests that relate to night driving performance, such as mesopic contrast sensitivity and glare assessment.

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